



Agentic Underwriting Copilot

An agentic reasoning, verification, policy & evaluation layer that turns a raw loan application into an evidence-backed, auditable recommendation for a human underwriter.

PRODUCTION READINESS REVIEW

Week 4.5 of 6 · Project 02

Team 02

Ayush Tiwary · Himkar Vashishta · Aditya Arora · Harshit Gautam

From docs-only to a benchmarked, auditable system

~60%

weighted readiness
(Eng + Eval up from ~16%)

0 vs 16

dangerous over-approvals
agentic vs single-LLM

1.61M

row real training table
Home Credit + LendingClub

16

automated tests green
safety invariants gated

The story, honestly

- In two sprints the project moved from docs-only to a runnable, safety-benchmarked, end-to-end slice on real data, taking the two heaviest, previously-empty buckets (Engineering 25% and Quality/Eval 25%) from near-zero to majority-done.
- The headline result: on a 128-case labelled golden set, the agentic pipeline catches 100% of regulatory hard-stops and commits 0 dangerous over-approvals, vs 26.5% and 16 for a single-LLM baseline, and grounds every decision with citations.
- We report real numbers, not a round headline. Every figure in this deck is reproducible from committed code. What's left, deploy, live LLM wiring, RAGAS/calibration, is scoped into the final 1.5 weeks.

We build the agentic layer, not the risk model

One raw application flows through a typed, checkpointed graph. Three worker agents run in parallel, a barrier merges them, the risk model scores, a synthesizer drafts a grounded recommendation, an evaluator gates it on faithfulness (retry ≤ 2), and a record-writer persists a replayable audit trail.



Agency is scoped to reasoning

- M = model decides (6 edges): the reasoning steps
- R = rule decides (5 edges): hard-stops, barriers, routing
- H = human decides (1 edge): the binding approve / decline

The model never disburses

The only irreversible, large-blast action, disbursing a loan, sits outside the agent, behind the human gate. The copilot advises; the underwriter decides. That single design choice is what makes the agency safe.

Locked the architecture, contracts & data

✓ ~95%

1

Architecture, sequence & state-schema diagrams

HLD / LLD / schema

2

Typed, schema-first shared state with write-isolation

state/application_state.py

3

8 agent contracts + failure-mode analysis

agents/nodes.py

4

Dataset defended vs 8 alternatives; North-Star = TDR

DATASET_COMPARISON.md

5

12 Architecture Decision Records

DECISIONS.md (ADR-001...012)

6

Repo scaffolding, CI stub, risk register, runbook

.github/ · docs/

Carried forward: LiteLLM gateway to live models · policy-corpus RAG spike.

One application, end-to-end, on real data

✓ ~70%



Runnable vertical slice

parallel → merge → risk → synth → eval → record



Audit replay

~10.7 recoverable checkpoints / decision



Real merged dataset, 1,613,898 rows

Home Credit + LendingClub • 0% feature nulls



Leakage guard + OOM-safe 1.2 GB reader

chunked allow-list load • ADR-012



Golden set, 128 cases, 8 strata

graded by an independent oracle



Single-LLM baseline + agentic benchmark

the headline result



Fairness audit on real data

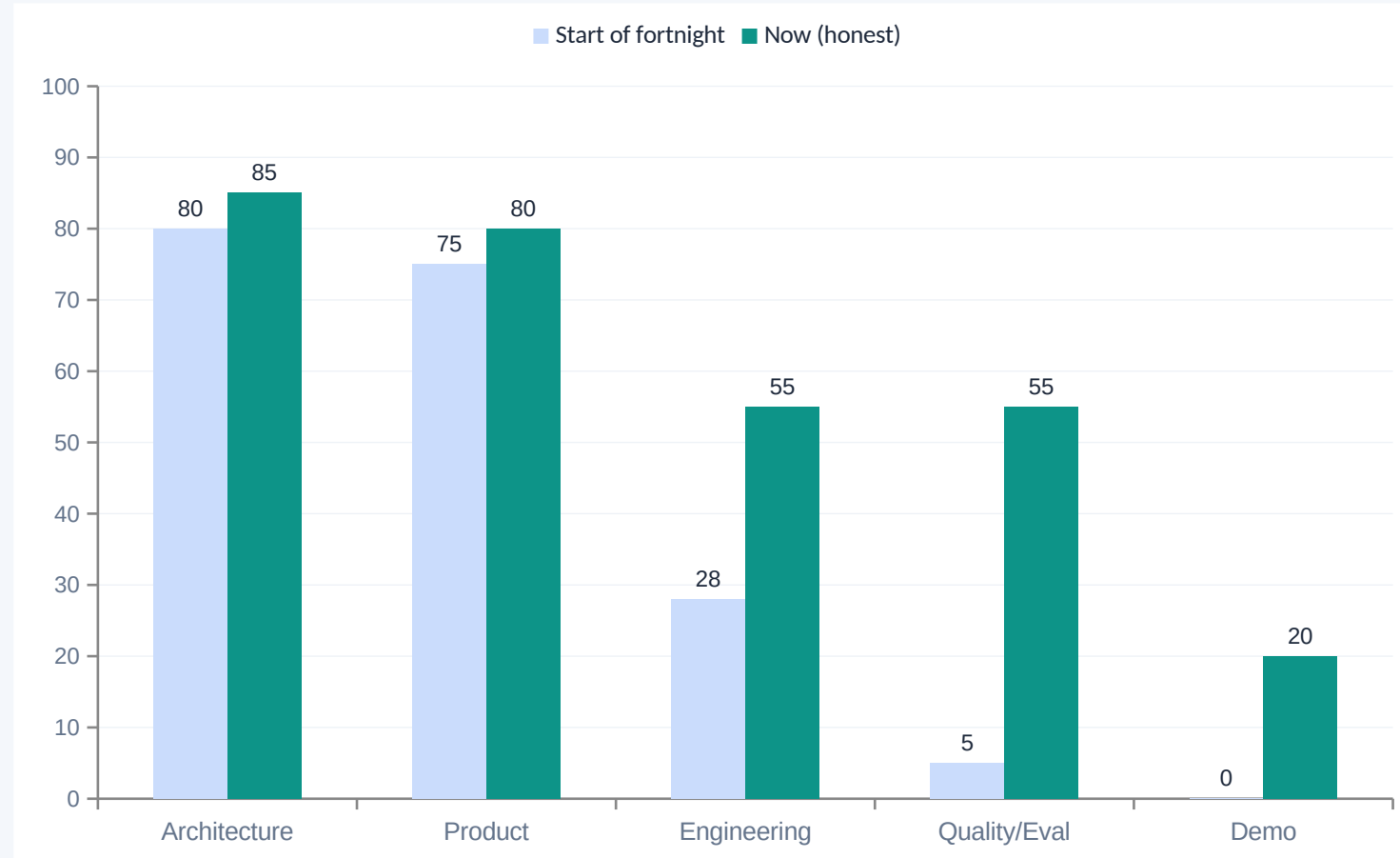
gender + age gaps surfaced



Cost / economics model + Readiness Scorecard

₹4.91 to ₹40.82 per decision

The two 25% buckets moved the most



Weighted readiness $\approx$ 60%

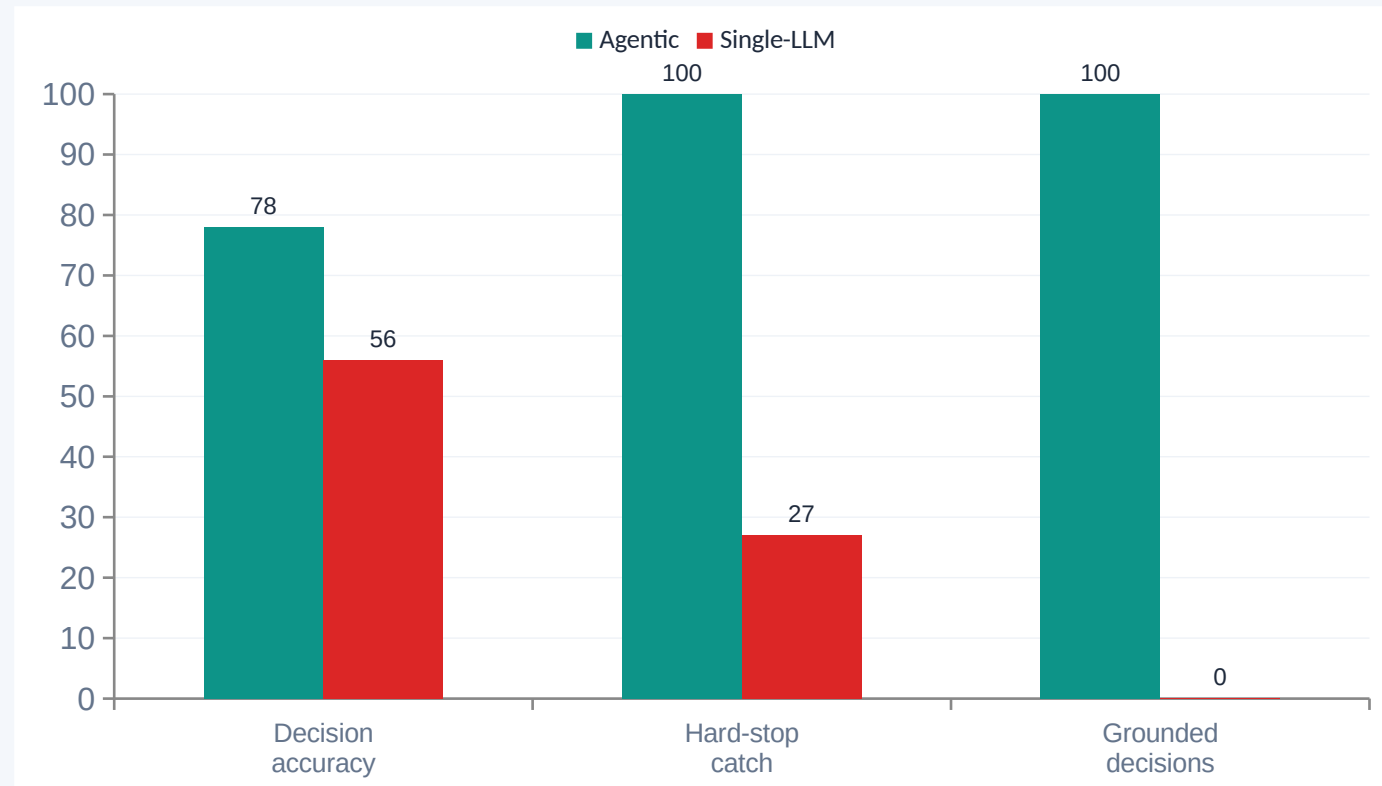
Path to 80% (final 1.5 weeks):

- LiteLLM live + real baseline numbers
- Policy-corpus ChromaDB RAG spike
- Deploy to free tier + record demo
- RAGAS runner + calibrated judge
- Production risk model + calibration

THE HEADLINE RESULT

Agentic vs. single-LLM baseline

128-case golden set (8 strata, seed 42), graded against an independent senior-underwriter oracle. Both engines are deterministic stand-ins, this isolates the structural safety the agentic design adds, before real LLM wiring.



0 vs 16

dangerous over-approvals

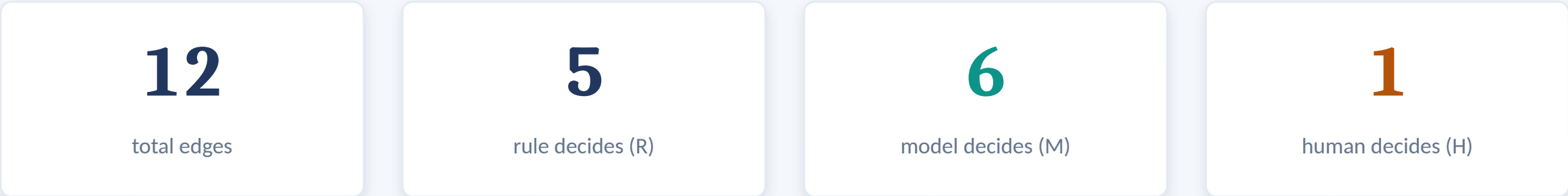
0 vs 36

hard-stops missed (of 49)

1.0

mean faithfulness (100% pass)

Agency, who decides at each edge

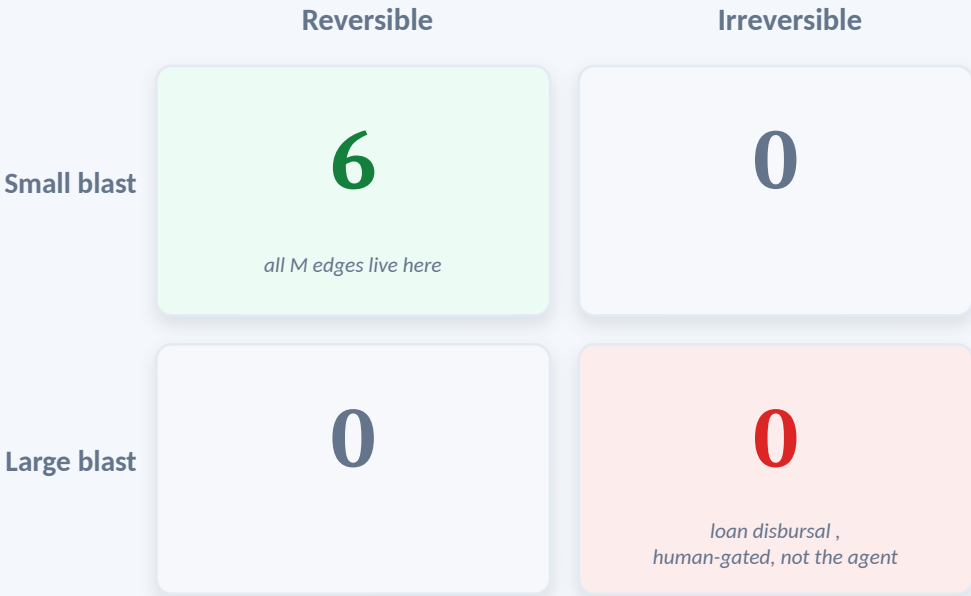


Agency present?	Yes
Scale of agency (M share)	50%, Medium (0 binding / irreversible edges)
Leash (longest unsupervised run)	3 boxes: risk → synthesize → evaluate (all checkpointed)
Riskiest M edge	“synthesize decision”, the approve/decline/refer draft

Verdict, does this problem deserve agency? Yes. Reconciling heterogeneous thin-file evidence and grounding it against policy is genuinely multi-step. But every model edge is bounded by typed state, mandatory citations, a faithfulness gate, deterministic hard-stops and a human on the binding call, agency is scoped to reasoning, never to irreversible action.

Before + After, blast radius × reversibility

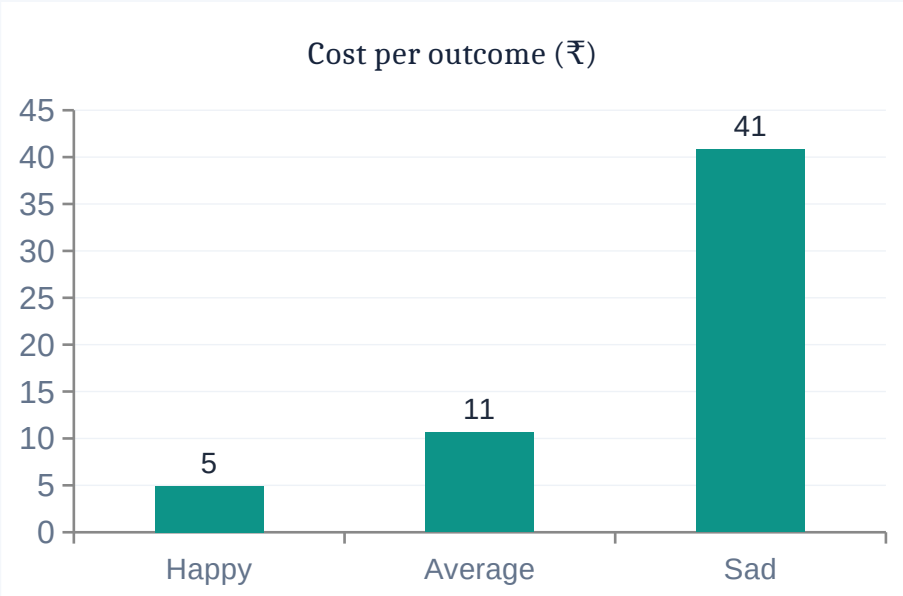
The agent's model edges, by risk quadrant:



Demo action table (excerpt)

M edge	Quadrant	What we built
synthesize decision	Rev+small (consequential)	Citations required + faithfulness gate + human sign-off → 0 over-approvals / 128
policy hard-stop	Irrev+large if acted on	Deterministic rule, not the model → 100% catch (0/49 missed)
risk band	Rev+small	Fairness measured on real data (DP 6.0pp; age 30→74%)
income/credit read	Rev+small	Verified (not stated) income + discrepancy flag

Economics, cost per loan decision



Input	Happy	Average	Sad
Outcomes / month	10,000	30,000	60,000
Model calls / outcome	5	5.4	9
Retries / outcome	0	0.2	2 (max)
H-gate trigger × min	8% × 3	15% × 4	40% × 6
Cost per outcome	₹4.91	₹10.63	₹40.82
Total monthly cost	₹49,064	₹3,18,865	₹24,49,089

Read: token cost is trivial (₹1 to 4 per decision), the human review gate dominates the bill. The cost lever is the escalation rate, which is exactly what the faithfulness gate controls. Prices: GPT-4o-mini + GPT-4o list rates, Jul 2026, FX ₹83/USD.

Business value, cheapest config that still holds

	Happy	Average	Sad
Total monthly cost	₹49,064	₹3,18,865	₹24,49,089
Faithfulness ≥ 0.80 (100% pass)	Y	Y	Y
Hard-stop catch = 100%	Y	Y	Y
0 dangerous over-approvals	Y	Y	Y
Decision accuracy ≥ 75% (78.1%)	Y	Y	Y
Fairness gap within threshold	measured	⚠	⚠
SLAs held (latency / availability)	Y	Y	N

Verdict

The Average configuration wins:

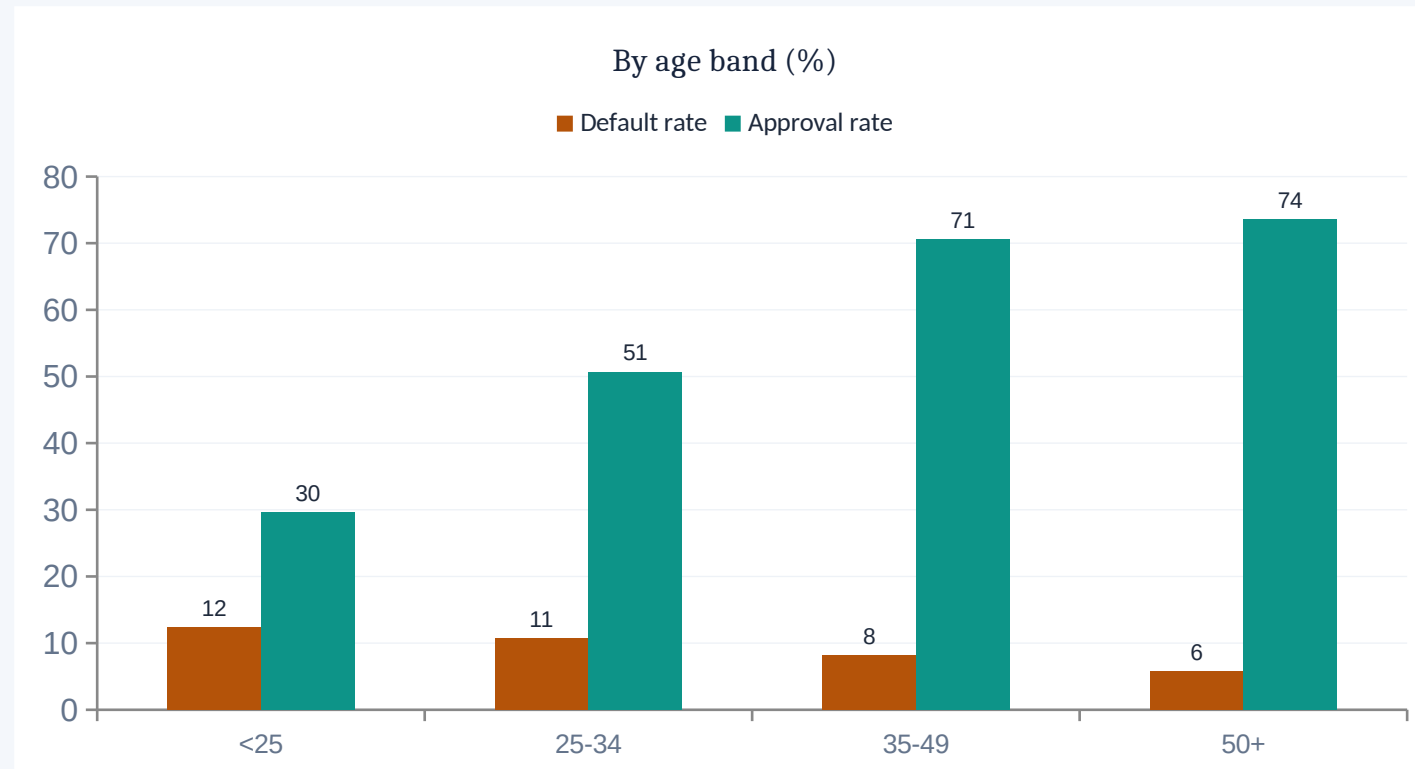
cheap/strong LiteLLM routing, retry cap 2, faithfulness gate 0.80, ~15% human gate.

All quality KPIs + SLAs met at ₹10.63 / decision (~₹3.2 L/month at 30k decisions).

The Sad path shows where to add an async human-review queue + fallback budget before scaling past ~60k/mo.

Fairness audit, on real data, not assumed

300,000 real Home-Credit applicants. A fairness-through-unawareness probe (protected attributes excluded from features) still inherits base-rate gaps, so we measure and surface them per decision rather than assume unawareness equals fairness.



Gender default

F 7.0% · M 10.1%

Demographic parity gap

6.0 pp (gender approval)

Equal-opportunity gap

4.9 pp (gender)

Age approval gradient

30% (<25) → 74% (50+)

What we delivered together this fortnight

1

Filled the two 25% buckets

Took Engineering + Quality/Eval from ~16% docs-only to a runnable, tested, benchmarked system on real data.

2

Produced the headline result

Agentic vs single-LLM, honest reproducible deltas, the project's defining artifact.

3

Ran a real fairness audit

300k applicants; concrete gender/age gaps surfaced, not assumed away.

4

Proved the unit economics

₹4.91 to ₹40.82 per decision; the human gate, not tokens, is the cost lever.

5

Completed the Readiness Scorecard

All five cards with numbers traceable to committed code.

6

Kept the repo the source of truth

12 ADRs, living PROGRESS board, CI-gated safety invariants.

One vertical each, every member defends their own



Ayush Tiwary

Evaluation, Policy-grounding & Architecture spine

Golden set + oracle, agentic-vs-baseline benchmark, faithfulness gate, fairness audit; ADR log + Master Build Plan; Readiness Scorecard.

“How do we know the agent is trustworthy, and how much better than one prompt?”



Himkar Vashishta

Model Gateway, Synthesis & Economics

LiteLLM cheap/strong routing, Decision Synthesizer contract, cost/economics model, single-LLM baseline; HLD/LLD/schema diagrams.

“What does each decision cost, and how is the model call bounded?”



Aditya Arora

Orchestration, State & Persistence

LangGraph graph (fan-out → barrier → retry loop), typed ApplicationState, DecisionRecord + audit replay, CI, forthcoming FastAPI.

“How does the graph run, recover, and produce a replayable audit trail?”



Harshit Gautam

Data & Risk Model

Dataset selection/EDA, HC+LC merge pipeline (canonical schema, adapters, leakage guard) → 1.6M-row table; risk kernel; PRD/TRD/research.

“Why this data, is it leakage-free, and how is risk scored and explained?”

Exactly what closes the gap, and what we'll demo

Path from ~60% → top grade

- 1 LiteLLM live + real single-LLM baseline numbers
- 2 Policy-corpus ChromaDB RAG spike (grounded retrieval)
- 3 Deploy to a free tier + record the 20-min demo
- 4 RAGAS runner + LLM-judge calibrated vs humans
- 5 Production risk model: GBM + SHAP + calibration curve
- 6 Fairness mitigation gate + CI regression gate

What we'll show tomorrow

- Live vertical slice, one application → decision + full trace + audit replay
- The benchmark, 100% vs 26.5% hard-stop catch; 0 vs 16 over-approvals
- The Readiness Scorecard, agency, blast-radius grid, ₹/decision economics
- The fairness audit, real gender/age gaps and how we surface them
- The honest roadmap, what closes the gap to a top-grade submission

From docs to a benchmarked, auditable system.

Every number in this deck is reproducible: `python -m eval.golden_set · eval.benchmark · eval.fairness · eval.economics`

Team 02 · Ayush Tiwary · Himkar Vashishta · Aditya Arora · Harshit Gautam

Halcyon Underwriting Copilot · Futureense × IIT Gandhinagar · Project 02